GAZA TRAINING CENTER

PROGRAMMING & DATABASE (PDB)

ONLINE TRAINING PLAN

ACADEMIC YEAR 2026–2027

Course title	Database Management
Course type	Specialized (PDB2)
Weekly schedule	3 periods per week | 14 teaching weeks
Credit hours	3
Instructor	Eng. Abeer Elaydi

Course Description

This course develops practical database-management skills for C# applications using Microsoft SQL Server. Students connect to SQL Server through ADO.NET, work with Entity Framework Core (EF Core), create reusable T-SQL database objects, apply transactions and permissions, inspect query performance, and prepare schema changes for deployment. Console applications support the practical work, while ASP.NET Core user-interface development remains part of the parallel Web course.

Educational Outcomes

- Explain the roles of SQL Server, ADO.NET, ORM, and EF Core in application data access.
- Build EF Core models, DbContext, LINQ queries, CRUD operations, and relationships.
- Develop T-SQL scripts, views, user-defined functions, and stored procedures.
- Apply transactions, database roles, permissions, and least-privilege principles.
- Interpret generated SQL and execution plans to improve basic query performance.
- Test and document database migrations for a database-driven application.

Course Objectives

01 Review relational database concepts and establish secure SQL Server connections from C#.

02 Configure and use EF Core for data mapping, querying, and schema changes.

03 Implement database programming with T-SQL control flow and reusable objects.

04 Manage data integrity using transactions and appropriate access permissions.

05 Investigate indexing, query efficiency, and deployment readiness.

Teaching Methods and Techniques

- Recorded Lectures: Pre-recorded explanations, coding demonstrations, and examples for self-paced learning.
- Live Online Practical Sessions: Instructor-guided coding, practical tasks, troubleshooting, and immediate feedback.
- In-Person Sessions Every Two Weeks: Face-to-face practice, discussion, problem-solving, and clarification.
- Independent Practice and Assessment: Coding exercises, assignments, quizzes, and formative feedback.

Tools and Platforms

Instruction and communication: Telegram, Google Classroom, and Microsoft Teams / Google Meet.
Practical implementation: Visual Studio or .NET CLI, SQL Server, SQL Server Management Studio (SSMS), and EF Core tools.

References

- Murach's SQL Server 2019 for Developers (2020).
- Smith, Jon P. Entity Framework Core in Action, 2nd ed. (2021).
- Gorman, Brian. Practical Entity Framework Core 10 (2026).
- Microsoft Learn: ADO.NET, EF Core, and Transact-SQL documentation.

Weekly Teaching Plan | Weeks 1-7

Chapter key: M = Murach (SQL Server 2019); G = Gorman (Practical EF Core 10); S = Smith (EF Core in Action, 2nd ed.); ML = Microsoft Learn.

Week	Main Topic	Content / Subtopics	References
1	SQL Server and ADO.NET	• SQL Server data-access architecture • SqlConnection, SqlCommand, and parameters • SqlDataReader and retrieving query results	M: Ch. 1 ML: ADO.NET
2	Introduction to EF Core	• ORM concepts and entity classes • DbContext, DbSet, and the SQL Server provider • Connection configuration and first queries	G: Ch. 1, 4 S: Ch. 1
3	EF Core CRUD Operations and Migrations	• Insert, retrieve, update, and delete records • SaveChanges() and change tracking • Code First migrations and schema updates	G: Ch. 3, 6 S: Ch. 3, 9
4	LINQ Queries and Asynchronous Operations	• Where, Select, and OrderBy • FirstOrDefault, Any, and Count • Async queries and query execution	G: Ch. 8-9 S: Ch. 2
5	Relationships and Related Data	• One-to-many and many-to-many relationships • Foreign keys and navigation properties • Eager and explicit loading of related data	G: Ch. 5 S: Ch. 8
6	T-SQL Scripting	• Variables and batches: DECLARE, SET, GO • Control flow: IF...ELSE, CASE, WHILE • Temporary tables	M: Ch. 14
7	Views and User-Defined Functions	• Create and alter SQL views • Scalar and table-valued functions • Reusable SQL Server database objects	M: Ch. 13, 15 G: Ch. 7

Weekly Teaching Plan | Weeks 8-16

Chapter key: M = Murach; G = Gorman; S = Smith; ML = Microsoft Learn. Selected readings support instructor preparation.

Week	Main Topic	Content / Subtopics	References
8	Stored Procedures and Midterm	• CREATE/ALTER PROCEDURE and EXEC • Input and output parameters • Midterm practical examination	M: Ch. 15 G: Ch. 7
9	Transactions and Error Handling	• ACID properties and transaction boundaries • BEGIN TRAN, COMMIT, ROLLBACK • TRY...CATCH and XACT_STATE	M: Ch. 16
10	Database Security	• SQL Server logins, database users, and roles • GRANT, REVOKE, and DENY • Principle of least privilege	M: Ch. 17
11	Indexes and Execution Plans	• Clustered and nonclustered indexes • Index seeks, scans, and selectivity • Basic query execution-plan analysis	M: Ch. 11 ML: Query plans
12	EF Core and T-SQL Integration	• Mapping database views in EF Core • Parameterized raw SQL queries • Using stored procedures with EF Core	G: Ch. 7 S: Ch. 11
13	Query Performance and Deployment Review	• Projection, pagination, and no-tracking • N+1 queries, split queries, and generated SQL • Review migration scripts and deployment checks	G: Ch. 8-9, 13 S: Ch. 9, 14-15
14	Integration, Revision and Project	• Integration review of EF Core and T-SQL concepts • Project demonstration and oral check • Revision and preparation for final assessment	Review: Weeks 1-13
15-16	Final Practical Examination	• Comprehensive final practical examination • Assessment of course competencies	Review: Weeks 1-14

Course Assessments

Marks	Evaluation Methods
30%	Midterm Practical Examination
30%	Final Practical Examination
20%	Quizzes, Assignments, Attendance and Engagement
10%	Integrated Database Project
10%	Oral Check / Demonstration / Evidence Submission
100%	Total

Course coordination: Database Management practical work uses Console applications for database exercises. ASP.NET Core user interface and HTTP development remain in the parallel Web course.